



Schedule Organizer

Luke Jeffries

Motivation



Finals week is
difficult to organize



Assignments vary in
workload



Due dates do not
reflect time required



**Goal: Help
students plan
time more
realistically**

Project Goal



Predict hours
needed from
assignment
descriptions



Support better
scheduling
decisions



Keep the system
simple and
interpretable

Dataset Overview



Artificially
generated
assignment
descriptions



Each task paired
with estimated
hours



No personal or
sensitive student
data used

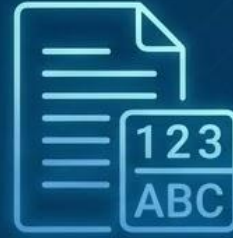
Feature Engineering



TF-IDF used to
convert text into
numerical features



Captures important
words and phrases



Added word count
and character count

Feature Extraction Implementation

Splits task text
into tokens

```
def featurize_task(task_text):  
    tokens = tokenize(task_text)  
    return {  
        "char_count": len(task_text),  
        "word_count": len(tokens),  
        "has_deadline": int("by" in tokens or "due" in tokens)  
    }
```

Constructs
feature vector

Checks for 'by' or
'due' keywords

Models Evaluated



Linear Regression
(baseline)



Random Forest
Regressor



Multi-Layer
Perceptron (MLP)



Random Forest selected for best performance

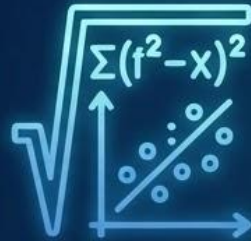
Models Evaluated (Part 2)

```
class WorkloadEstimator:
    def __init__(self):
        self.pipeline = Pipeline([
            ("tfidf", TfidfVectorizer(ngram_range=(1,2))),
            ("clf", RandomForestRegressor(n_estimators=200))
        ])
    def fit(self, X, y): self.pipeline.fit(X, y)
```


Evaluation Results



Mean Absolute
Error (MAE)

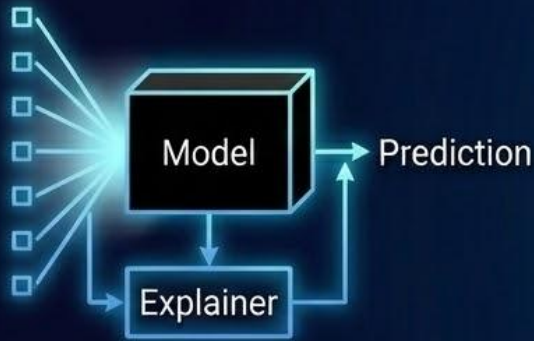


Root Mean Squared
Error (RMSE)



Cross-validation
used for reliability

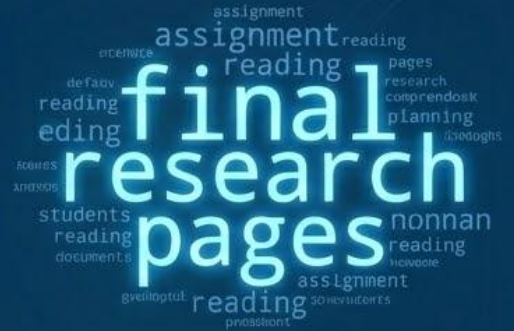
Explainability (SHAP)



SHAP explains model predictions



Shows which words
increase estimated time



Important features:
final
final, research, pages

SHAP Code Implementation

```
explainer = shap.TreeExplainer(trained_model.pipeline.named_steps["clf"])  
shap_values = explainer.shap_values(tfidf_transformer.transform(texts))  
shap.summary_plot(shap_values, feature_names=tfidf_vectorizer.get_feature_names_out())
```


Limitations & Future Work



Limitations



Not
personalized
yet



Uses synthetic
data



Future Work



Learn from
user habits



Calendar export
(Google/iCal)

Conclusion



AI can estimate
assignment
workload



Interpretability
builds trust



Strong foundation
for smart
scheduling tools